BENHA NATIONAL UNIVERSITY
MECHATRONICS ENGINEERING

6DOF ROBOT ARM
FORWARD KINEMATICS REPORT

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Date: 2026-06-13

1. Objective

This report analyzes forward kinematics of a 6DOF robotic arm using Denavit–Hartenberg parameters and transformation matrices.

2. Joint Angles (θ)

Joint	Name	Angle (deg)
θ_1	Base (NEMA 23)	80
θ_2	Shoulder (Servo 35)	50
θ_3	Elbow (Servo 35)	40
θ_4	Rest (Servo 11/20)	0.0
θ_5	Roll (NEMA 17)	0.0
θ_6	Yaw (Servo 11)	0.0
θ_7	Gripper(Black Servo)	0.0

3. Link Dimensions

Parameter	Value (mm)
a2 (Shoulder)	150.0
a3 (Elbow)	130.0
d1 (Base)	100
d4 (Wrist Pitch)	74.17
d5 (Wrist Roll)	71.09
d6 (Gripper)	40

4. Denavit–Hartenberg Parameters

Link	θ	d	a	α
1	80	100	0	90
2	50	0	150.0	0
3	40	0	130.0	0
4	0.0	74.17	0	90
5	0.0	71.09	0	-90
6	0.0	40	0	0

5. Transformation Matrices (A1 → A6)

A1			
0.17	-0.00	0.98	0.00
0.98	0.00	-0.17	0.00
0.00	1.00	0.00	100.00
0.00	0.00	0.00	1.00

A2			
0.64	-0.77	0.00	96.42
0.77	0.64	-0.00	114.91
0.00	0.00	1.00	0.00
0.00	0.00	0.00	1.00

A3			
0.77	-0.64	0.00	99.59
0.64	0.77	-0.00	83.56
0.00	0.00	1.00	0.00
0.00	0.00	0.00	1.00

A4			
1.00	-0.00	0.00	0.00
0.00	0.00	-1.00	0.00
0.00	1.00	0.00	74.17
0.00	0.00	0.00	1.00

A5			
1.00	-0.00	-0.00	0.00
0.00	0.00	1.00	0.00
0.00	-1.00	0.00	71.09
0.00	0.00	0.00	1.00

A6			
1.00	-0.00	0.00	0.00
0.00	1.00	-0.00	0.00
0.00	0.00	1.00	40.00
0.00	0.00	0.00	1.00

6. Final Transformation Matrix (T06)

A6			
1.00	-0.00	0.00	0.00
0.00	1.00	-0.00	0.00
0.00	0.00	1.00	40.00
0.00	0.00	0.00	1.00

7. End Effector Position

X	141.52
Y	145.14
Z	344.91

8. Assignment Questions

1: What is the purpose of Forward Kinematics in robotic arms?

2. What is the physical meaning of Denavit–Hartenberg parameters?

3. How does link length (a2, a3) affect end-effector position?

4. What is the difference between rotation matrix and transformation matrix?

5. Why do we multiply transformation matrices in FK?

6. What happens to end-effector position if joint angles change?

7. Explain the importance of homogeneous transformation in robotics.

8. How does changing d parameters affect robot workspace?